

Smart Utility Sustainability Dashboard

4bn

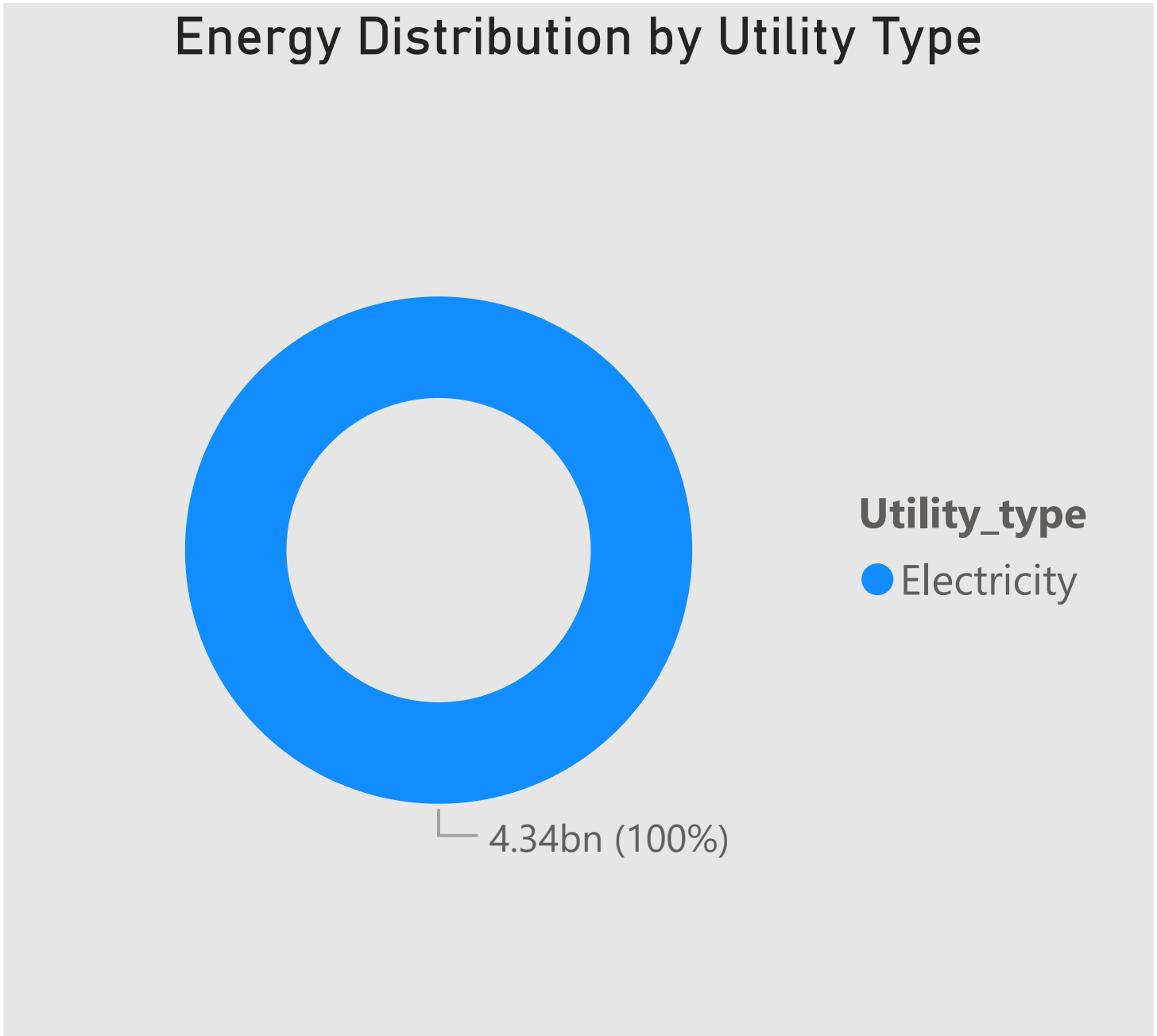
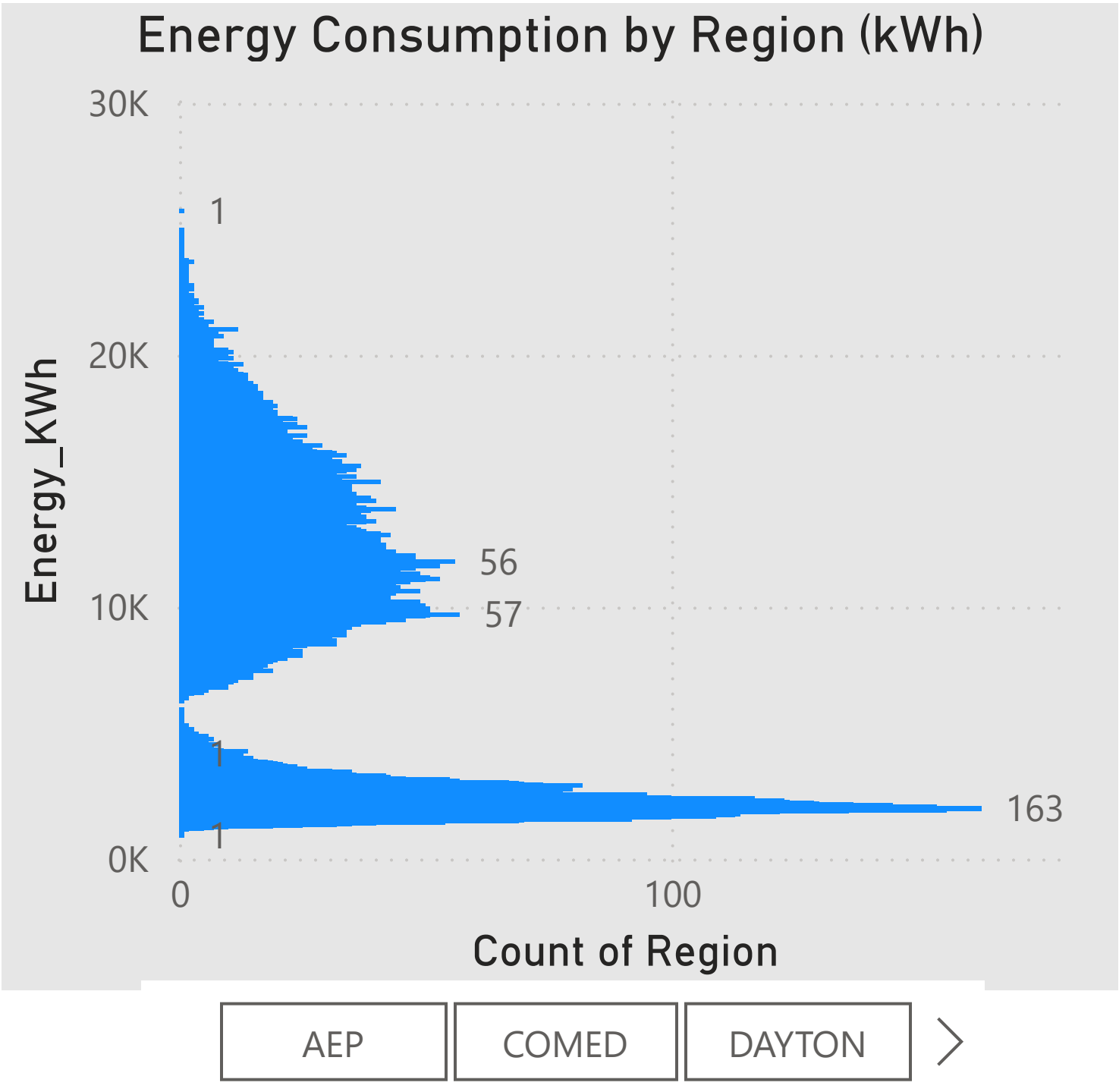
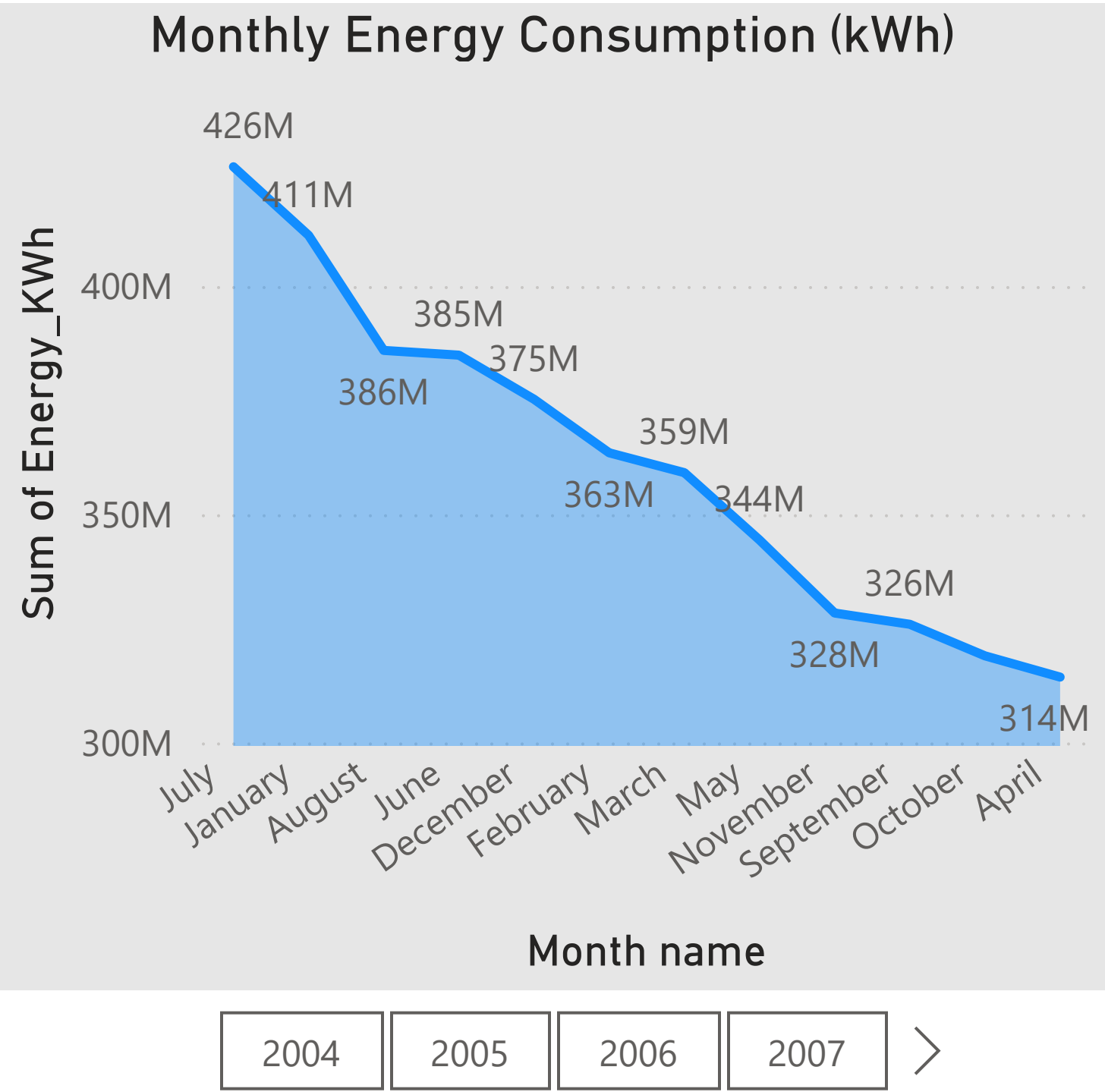
Sum of Energy_KWh

8.98K

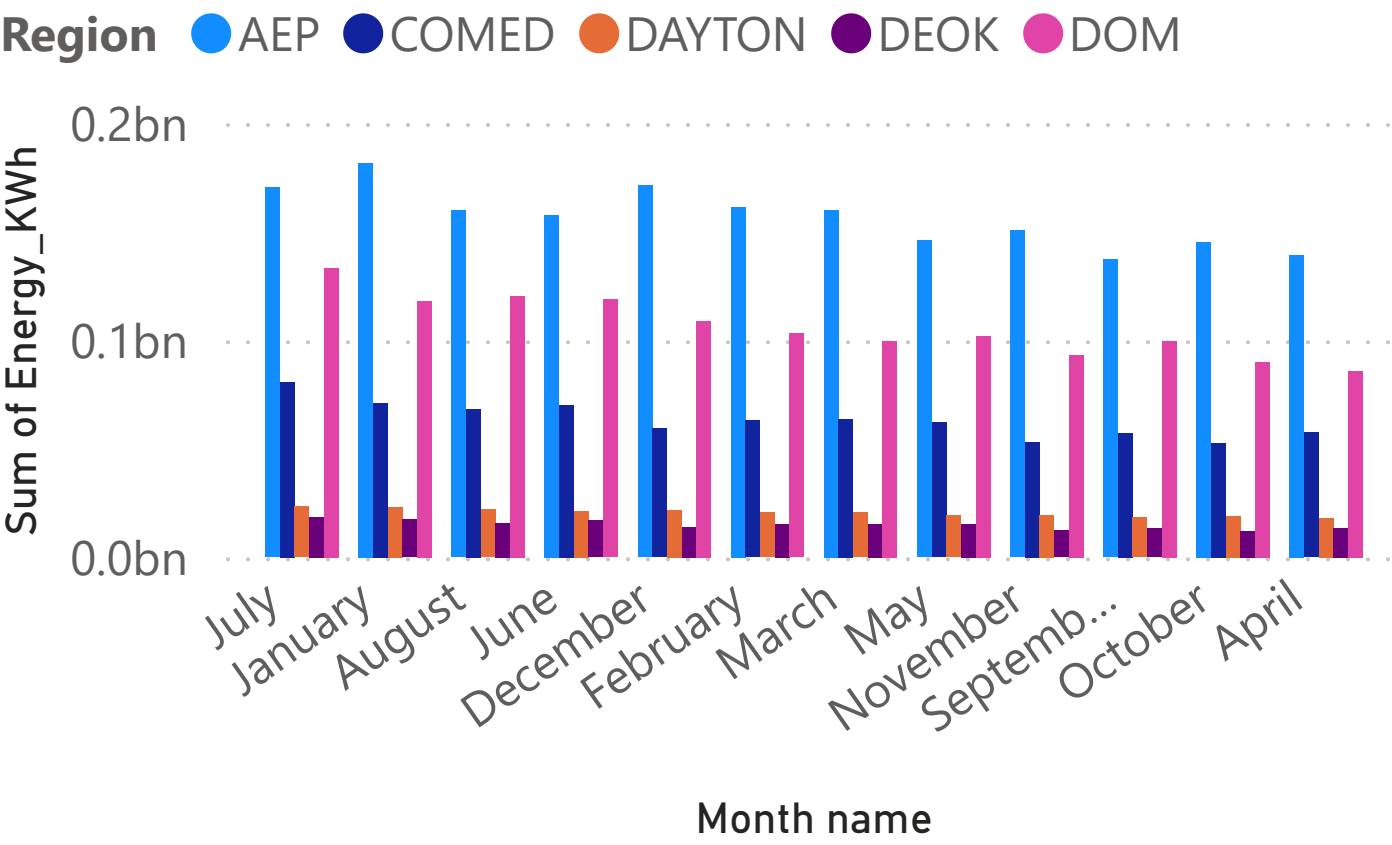
Average of Energy_KWh

5

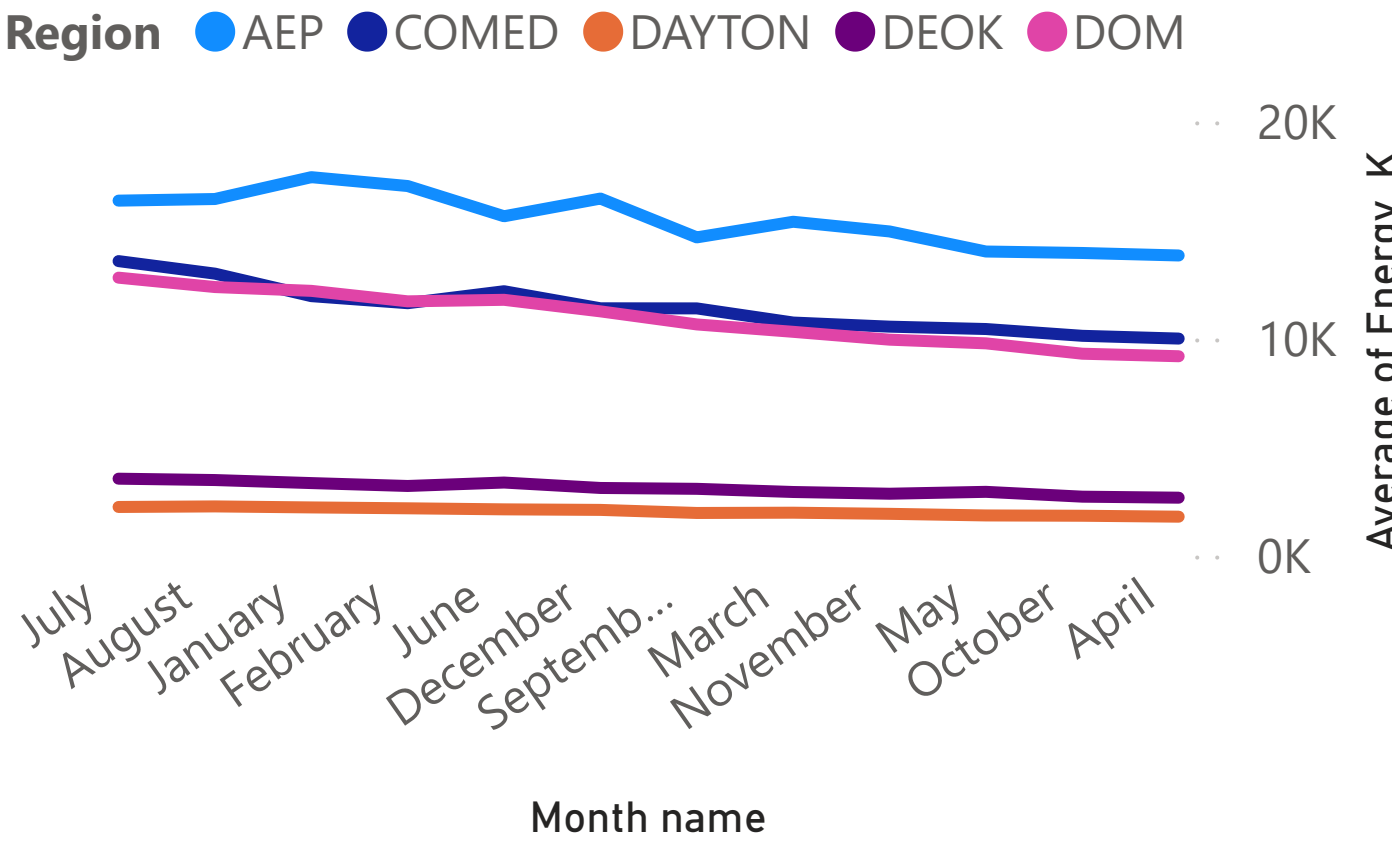
Count of Region



Monthly Energy Consumption (kWh)



Energy Trend (Line View)



Total Consumption – Highest Region

1.88bn

Total Consumption – Lowest Region

179.29M

Peak Month – Total Energy (kWh)

426.13M

Lowest Month – Total Energy (kWh)

314.32M

Regional & Seasonal Energy Insights

Comparing total usage, seasonal peaks, and region rankings

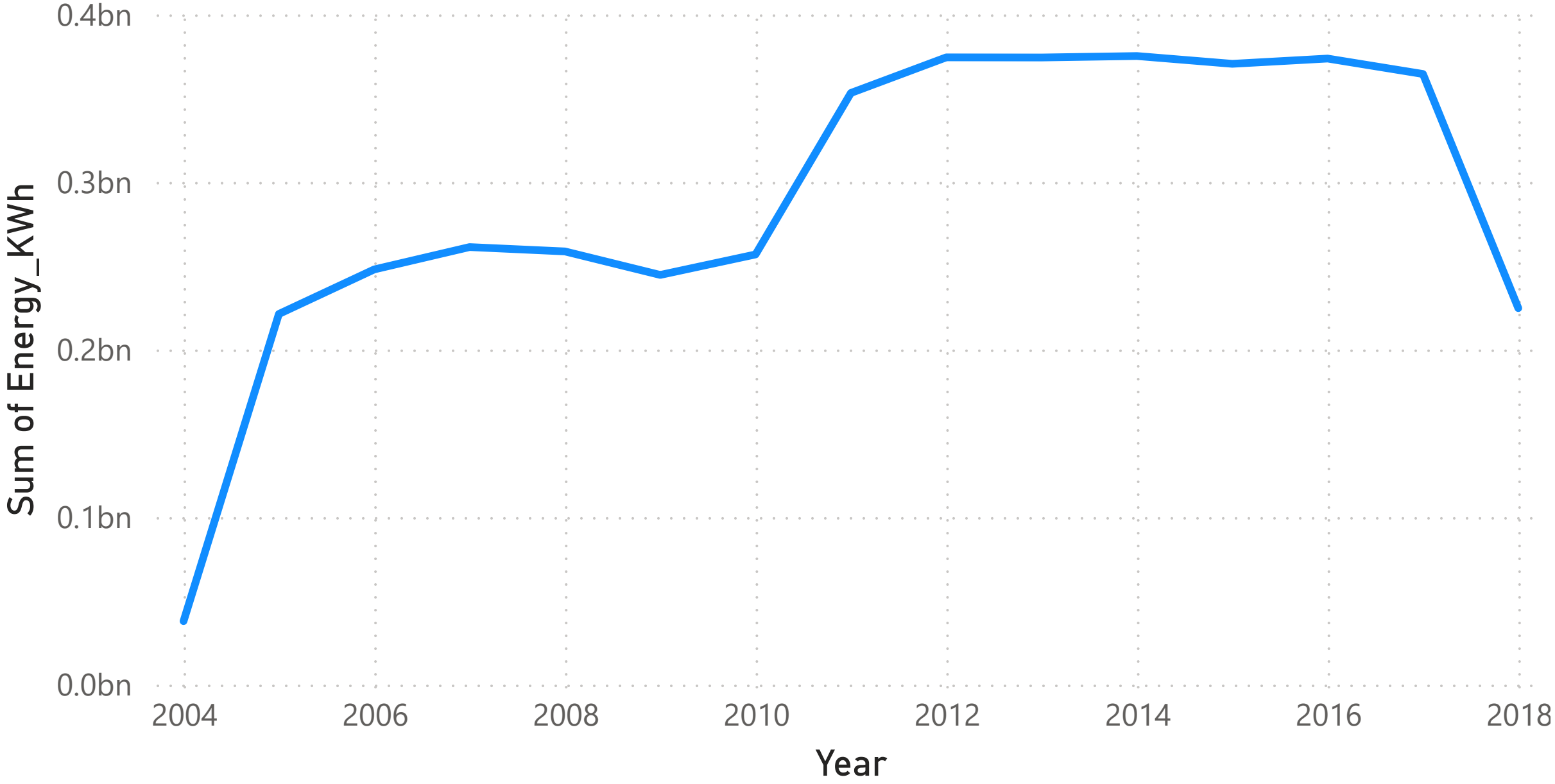
Region	Sum of Energy_KWh
AEP	1879672527
DOM	1272177020
COMED	759405855
DAYTON	247140397
DEOK	179285166

Load Profiles & Multi-Year Analysis															
Region	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
AEP	74032471	70207532	67805054	66447607	66170012	66917668	69757337	74706552	78230008	79967692	81289971	82412119	82876387	83278766	83549638
COMED	31234298	29060764	27474324	26362729	25783454	25535170	25902244	27200635	29311863	30922650	32059346	32956390	33746465	34195950	34468328
DAYTON	9609331	9030503	8636539	8387879	8307620	8405010	8843445	9634267	10213629	10501299	10775813	11009130	11110653	11200895	11244237
DEOK	7021965	6603515	6330620	6139917	6081556	6138681	6427543	6925963	7270609	7469606	7669581	7864107	7972988	8055933	8117841
DOM	48548515	45477655	43626707	42626778	42498532	43295766	45849346	49882071	52574189	53866867	54853097	55774330	56412619	56856810	57226662
Total	170446580	160379969	153873244	149964910	148841174	150292295	156779915	168349488	177600298	182728114	186647808	190016076	192119112	193588354	194606706



Energy Trend by Utility Type

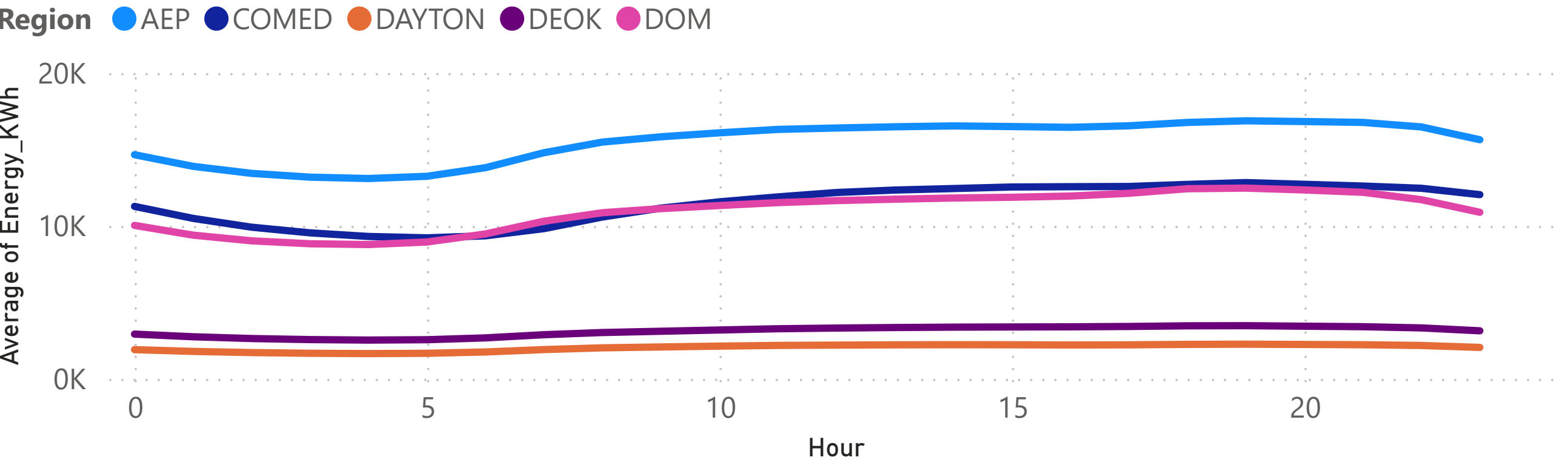
Utility_type ● Electricity



Region performance comparison table

Region	Sum of Energy_KWh	Average of Energy_KWh	Min of Energy_KWh	Max of Energy_KWh
AEP	1879672527	15499.513716985644	9581	25695
COMED	759405855	11420.152112125359	7237	23753
DOM	1272177020	10949.203625128024	1253	21651
DAYTON	247140397	2037.85113997114	982	3746
Total	4337680965	8981.20798678187	907	25695

Hourly Load Profile (Line Chart)



Executive Summary & Insights Page

Executive Summary:

This report analyzes hourly and monthly electricity consumption across the five major utility regions (AEP, COMED, DAYTON, DEOK, DOM) from 2004–2018. The dashboard highlights regional performance, seasonal patterns, and annual consumption trends to support operational planning and sustainability initiatives.

Key Findings

- **AEP is the highest-consuming region** with **1.87B kWh** total usage.
- **DEOK is the lowest-consuming region** with **179M kWh** usage.
- **Peak month is January** with approx **426M kWh**, likely due to winter heating loads.
- **Lowest month is April** with approx **314M kWh**, tied to mild weather.
- **Overall energy consumption peaks between 2011–2014**, indicating growth in load demand.

Operational Insights

- Regions with the highest consumption (AEP, COMED) may require **grid reinforcement** and **capacity planning**.
- Winter and summer peaks suggest **seasonal forecasting models** can optimize generation.
- Load stability in low-demand months provides opportunity for **maintenance scheduling**.

Recommendations

- Invest in **load balancing and smart grid technologies** in AEP & COMED.
- Develop **energy reduction programs** during winter peaks.
- Explore **renewable integration** during off-peak seasons.
- Improve **demand forecasting** using historical seasonal patterns.

Years of Data (2004–2018)

15

Total Data Points Analyzed

482.97K

Overall Average Daily Energy Usage (kWh)

8.98K